

Jonathan Zhu

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SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL

Backend / Systems: FastAPI, Node.js, Express.js, REST APIs, JWT, OAuth2, Pydantic, SQLAlchemy

Frontend: React, Next.js, TypeScript, Tailwind CSS, Material UI, React Query, Framer Motion

Databases: PostgreSQL, MySQL, MongoDB

Cloud / Infrastructure: AWS (EC2, RDS, S3, Bedrock, Cognito, API Gateway, Amplify, Lambda, Secrets Manager, VPC), Terraform

AI / ML: Amazon Nova Pro, Anthropic API, OpenAI API, Gemini API, Prompt Engineering, Multimodal LLMs

Tools & Compliance: Jenkins, Splunk, Dynatrace, HIPAA Compliance, BAA, Docker

WORK EXPERIENCE

Citywide Eye Care

Great Neck, NY

Full-Stack Software Engineer

Jan 2024 – Present

- Lead engineer for a production EMR platform used daily by clinical staff, architecting a zero-downtime migration from on-premise servers to AWS (EC2, RDS, S3, Cognito, API Gateway, Amplify)
- Owned HIPAA security architecture end to end: Cognito JWT auth, IP whitelisting, role-based access control, and WireGuard VPN, passing all external compliance reviews with zero audit findings
- Reduced physician documentation time by 75% by engineering structured charting workflows with auto-save, copy-forward, and S3-backed document storage via pre-signed URLs
- Partnered directly with clinicians to define requirements and iterate on features, accelerating feedback cycles and cutting rework across all major releases
- Architected for high availability and fault tolerance on AWS using VPC-isolated services, S3 redundant document storage, and Multi-AZ RDS with automated backups ensuring 24/7 uptime for clinical operations

ZLA - AI-Enabled CRM Platform

Rochester, NY

Full-Stack Software Engineer

Aug 2025 – May 2026

- Built a production CRM from scratch on a 5-person Agile team, delivering lead management, property tracking, campaign workflows, and agent dashboards within tight contract timelines
- Engineered an AI-powered lead discovery pipeline integrating Google Places API, Geocoding, Brave Search, and OpenAI, cutting manual prospect research time by 60%
- Architected a normalized FastAPI and PostgreSQL backend with complex relational schemas; built a React/TypeScript frontend with Kanban pipelines and map-based lead search
- Provisioned and managed AWS infrastructure with Terraform (Amplify, EC2 with Elastic IP, API Gateway, Lambda) and built an admin dashboard for one-click backend server management

Citywide Eye Care

Great Neck, NY

Software Engineer Intern

Sep 2023 – Dec 2023

- Built and deployed an AI health chatbot using Python and Gemini API, autonomously handling patient FAQs and symptom triage, cutting front-desk call volume by 50%
- Rebuilt the patient-facing website in Next.js and Material UI, improving SEO and driving a 25% increase in appointment conversion rates
- Implemented digital intake and scheduling workflows on AWS (EC2, S3, RDS, Secrets Manager), replacing all manual paper-based processes

ADP

Roseland, NJ

Site Reliability Engineer Intern

Jun 2023 – Aug 2023

- Developed an automated service health and status-reporting tool integrated with Splunk and Dynatrace, eliminating manual monitoring workflows and reducing incident response turnaround time by 20%
- Improved deployment reliability and reduced release cycle time by 30% by optimizing Jenkins CI/CD pipelines and tuning MySQL query performance across internal analytics services
- Owned observability integrations across S3, Splunk, and Dynatrace to surface system-level metrics, logs, and alerts; recognized by engineering leadership for measurable impact on operational visibility and team velocity

PROJECTS

- RemitAI (github.com/Jonathanz4344/RemitAI):** HIPAA-compliant AI platform for healthcare RCM using Amazon Nova Pro on AWS Bedrock (BAA); processes EOB/ERA PDFs via multimodal pipeline, classifies denials by CARC/RARC codes, generates appeal strategies, and builds a self-learning payer knowledge base on async FastAPI, PostgreSQL, and React/TypeScript
- AI Lead Discovery Pipeline (github.com/Jonathanz4344/BigRealEstate):** Multi-source enrichment pipeline integrating Google Places API, Geocoding, Brave Search, and OpenAI to surface qualified real estate prospects, cutting manual research time by 60%
- Enlighten Learning (enlightenlearning.vercel.app):** Full-stack nonprofit site built in React, TypeScript, and Material UI with serverless API integration (Google Apps Script), dark/light mode, animated UI, and donation tracking; supports tutoring programs for students in underserved communities across Long Island

EDUCATION

Georgia Institute of Technology

Expected May 2028

M.S. in Computer Science

Rochester Institute of Technology

May 2026

B.S. in Software Engineering